Supplemental Online Content

Embleton ND, Sproat T, Uthaya S, et al. Effect of an exclusive human milk diet on the gut microbiome in preterm infants: a randomized clinical trial. *JAMA Netw Open.* 2023;6(3):e231165. doi:10.1001/jamanetworkopen.2023.1165

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This supplemental material has been provided by the authors to give readers additional information about their work.

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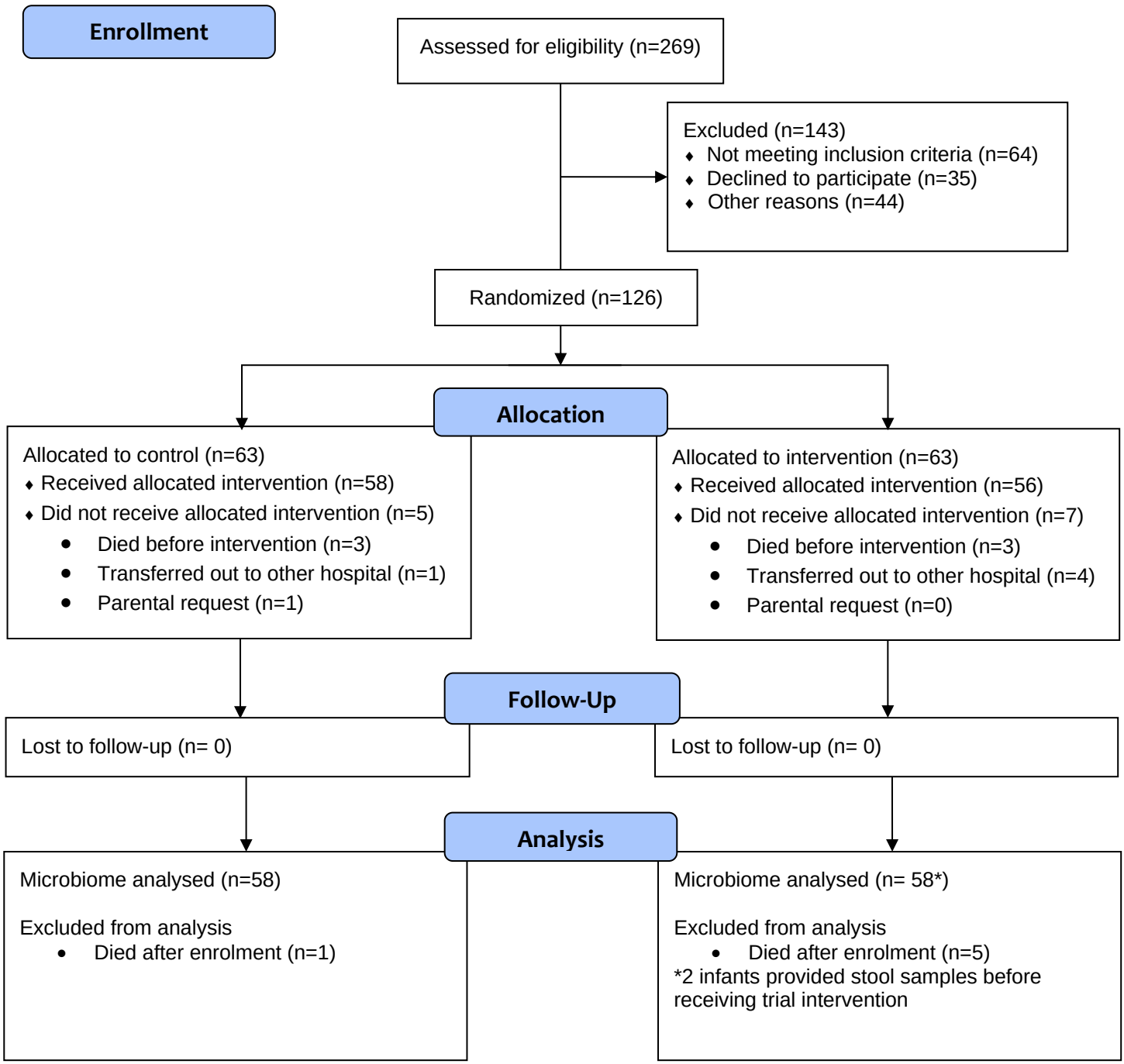
eFigure1. CONSORT Flow Diagram



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eTable 1. Causes of Death in Recruited Infants

Study arm	Gestation (weeks), Birthweight (closest 100g)	Age at death (days)	Days of MOM exposure	Days of formula or RTF	Days of fortifier	Primary cause of death
Intervention	27, 500	12	8	8	0	Respiratory failure
Intervention	24, 600	3	0	0	0	Intraventricular haemorrhage
Intervention	24, 500	19	1	15	0	Respiratory failure
Intervention	25, 800	6	3	2	0	Late onset sepsis (Enterococcus spp.)
Intervention	25, 900	10	4	7	0	Congenital anomaly*
Control	28, 500	133	0	117	0	Chronic lung disease**
Intervention	26, 900	15	9	0	0	Intraventricular haemorrhage**
Control	23, 500	34	23	0	0	Late onset sepsis (*Staph aureus*)
Intervention	24, 500	32	29	0	0	Congenital Herpes Simplex Virus infection
Control	24, 700	12	0	0	0	Fungal infection
Control	25, 800	15	7	1	0	Respiratory failure
Intervention	26, 400	112	107	14	37	RSV infection

*Footnote: birthweight rounded to nearest 100g to maintain anonymity, *congenital anomaly not recognised until after recruitment **developed surgical NEC but not primary cause of death*

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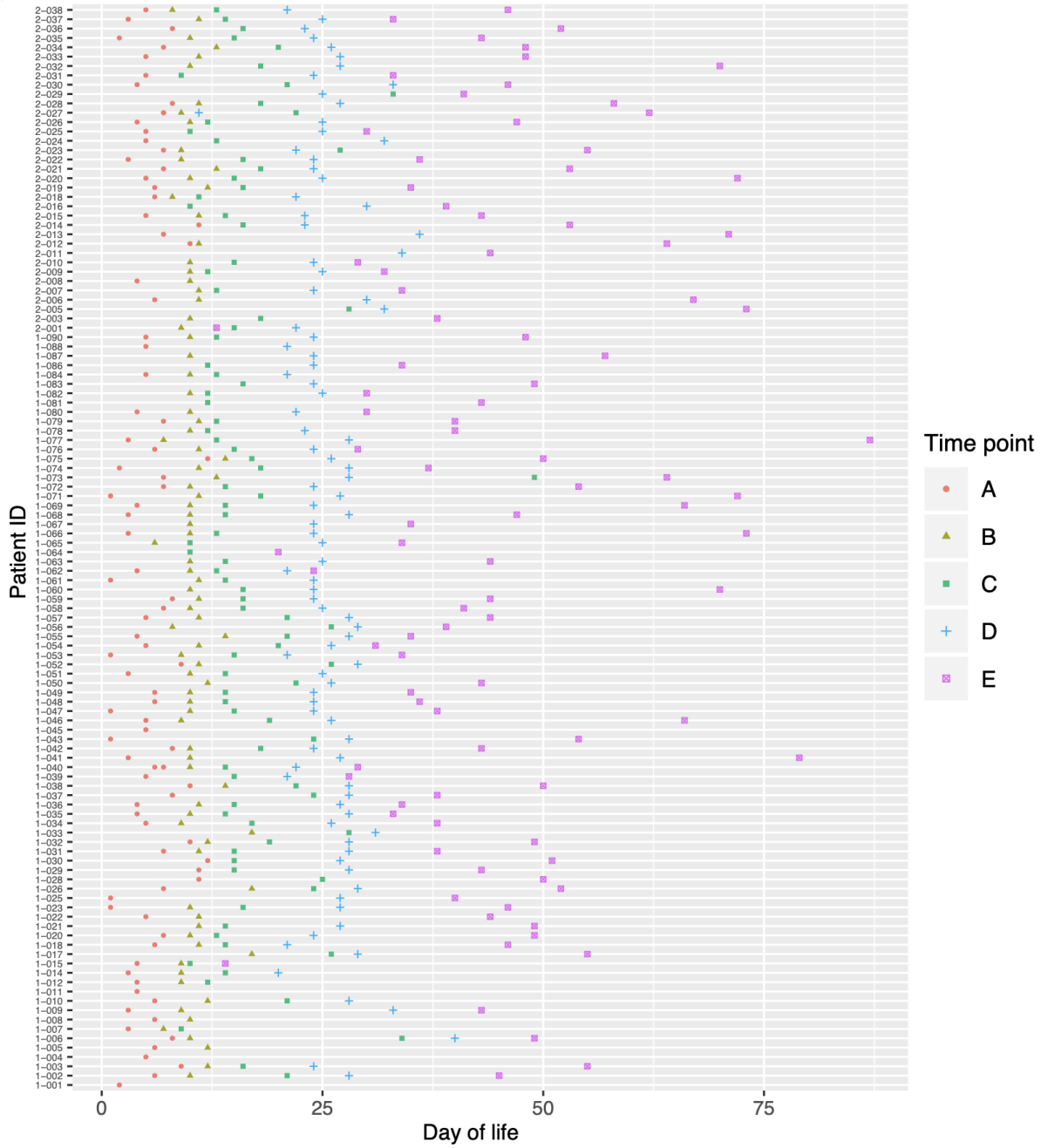
eFigure 2. Schematic showing samples included from each infant per day of life



Footnote. Samples are coloured and shaped according to time point.

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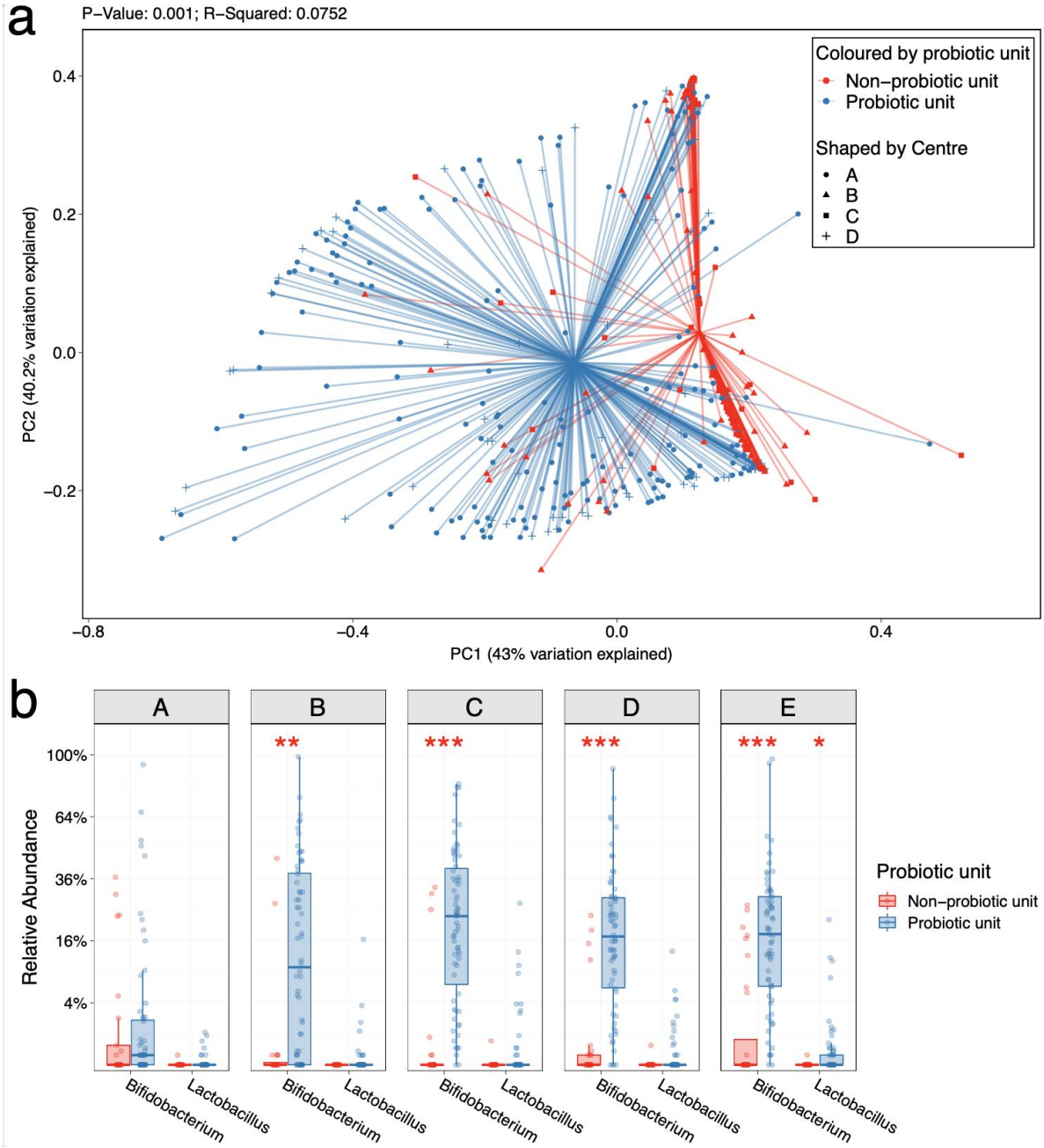
eFigure 3. Importance of probiotic use in shaping the preterm gut microbiome.



Footnote. a, Weighted UniFrac PCoA of all samples. Samples are coloured by probiotic use on the unit and shaped by the centre. b, Box plots showing the relative abundance of Bifidobacterium and Lactobacillus. These species are used in preterm probiotics and were the only genera associated with probiotic use. * denotes P < 0.05, ** denotes P < 0.01, *** denotes P < 0.001.

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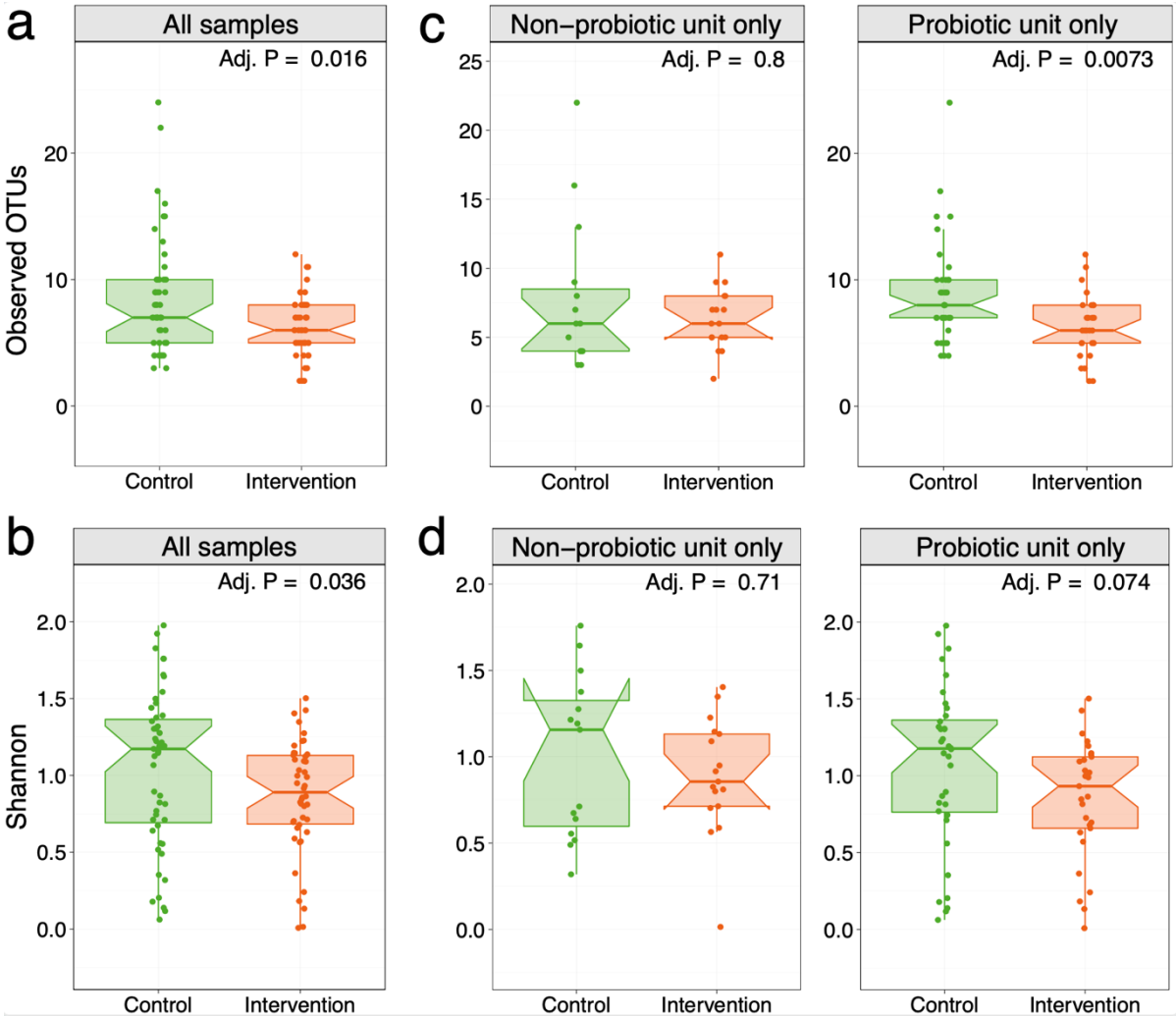
eFigure 4. Boxplots of Shannon diversity at time point E (study completion) stratified by probiotic use



Footnote. Boxplots of the Shannon diversity at time point E (study completion) showing all samples and samples stratified by probiotic use on the NICU. a, Number of observed OTUs. b, Shannon diversity .

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eTable 2. MaAsLin2 results for significant taxa at the genus level

Covariate	Genus	Variable	Coefficient	Stderr	P value	Q value
Birthweight	Streptococcus	Birthweight	0.01602964	0.00685795	0.02145566	0.32288038
Birthweight	Haemophilus	Birthweight	0.01180561	0.00530623	0.02802357	0.36430636
Days_MOM	Bifidobacterium	Days_MOM	0.07521571	0.02958338	0.01246316	0.25923368
DOL	Staphylococcus	DOL	-0.1728678	0.01714974	2.38E-21	4.95E-19
DOL	Corynebacterium	DOL	-0.006291	0.00245029	0.01061696	0.2453697
DOL	Enterobacter	DOL	0.12486038	0.01817296	2.69E-11	2.80E-09
DOL	Veillonella	DOL	0.03383581	0.00581765	1.26E-08	8.74E-07
DOL	Clostridium_sensu_stricto_1	DOL	0.0183702	0.00341432	1.26E-07	6.57E-06
DOL	Escherichia_Shigella	DOL	0.04976307	0.01356903	0.00028164	0.00976367
DOL	Bifidobacterium	DOL	0.04196848	0.01207922	0.00056827	0.0168858
Mode_delivery	Lactobacillus	CS	0.05590385	0.02144088	0.01042481	0.2453697
MOM_at_34_weeks	Clostridium_sensu_stricto_1	Y	-0.0340969	0.01415225	0.01788234	0.32288038
Probiotic_unit	Bifidobacterium	Probiotic Unit	0.2853622	0.05809966	3.48E-06	0.00014484
Sex	Enterobacter	Male	0.17673536	0.07578834	0.02173233	0.32288038
Total_days_Antibiotics	Bifidobacterium	Total_days_Antibiotics	-0.0439683	0.01885785	0.02141536	0.32288038
Trial_arm	Lactobacillus	Intervention	-0.017354	0.00766966	0.02572098	0.35666427

Footnote. Mixed-effects linear models using a variance-stabilizing arcsin square root transformation on relative abundance genera data were used to determine the significance. Top 15 most significant associations are shown. Results adjusted for covariates and controlling for repeated measures

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